INFOSYS SE ASSESSMENT 2026 • OFFICIAL PREPARATION VAULT

Infosys SE Complete Preparation — Number Decimal & Fractions

Section: prepinsta infosys SE | **Drive Date:** 5th & 6th November 2026 | **Status:** 100% Solved & Verified

Infosys SE Complete Preparation — Number Decimal & Fractions

Source Portal: PrepInsta Infosys Preparation Suite (`https://prepinsta.com/infosys/aptitude/number-decimal-fractions/`)
Target Role: Infosys Systems Engineer (SE) | **Category:** 01_Mathematical_Ability_Technical_Aptitude
Total Verified Questions Extracted: 10
Quizzes Covered: quiz-1, quiz-2, quiz-3, quiz-4, quiz-5, quiz-6

📋 Section 1: Rules to Keep in Mind

These rules and fundamental principles govern all Infosys questions on this topic:

- BODMAS / PEMDAS operator priority rule.
- Converting recurring decimals to vulgar fractions: 0.bar(a) = a/9; 0.bar(ab) = ab/99; 0.a bar(b) = (ab - a)/90.
- Comparing fractions: Cross-multiply numerators and denominators to determine relative size.

⚡ Section 2: How to Solve Quickly

Proven time-saving strategies recommended by PrepInsta for Infosys 2026 examination:

1. **Option Elimination via Unit Digits & Multiples:** Before doing full calculations, inspect the options to see if last digits or divisibility by 3, 7, 9, 11 can eliminate 2 to 3 wrong choices.
2. **Approximation & Magnitude Checks:** When exact arithmetic is tedious, round numbers to neat multiples (e.g., 99 ≈ 100) to find the nearest valid option.
3. **Reverse Option Substitution:** In complex algebraic equations, plug option values directly into the question stem to test validity.

🚀 Section 3: Tricks & Shortcuts

Key mathematical formulas, mental-math shortcuts, and speed axioms:

- Decimal multiplication: Multiply ignoring decimals, then place decimal point equal to total sum of decimal places.
- If numerator and denominator have same difference, fraction with larger numbers is greater.

📚 Section 4: Practice Questions, Options & Step-by-Step Solutions

Contains **10** verbatim questions with options, verified answer keys, and complete derivations:

Question 1 (PrepInsta quiz-1 - Q1)

Question Stem:
A tailor has 37.5 metres of cloth and he has to make 8 pieces out of a metre of cloth. How many pieces can he make out this cloth ?

Options:
- **(A)** `300`
- **(B)** `360`
- **(C)** `400`
- **(D)** `450`

Correct Answer: (A) `300`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Length of each piece = (1/8) m = 0.125 m


Required no of pieces = (37.5/0.125) = (375 * 100) / 125 = 300.
```

Question 2 (PrepInsta quiz-1 - Q2)

Question Stem:
If 47. 2506 = 4 \times A + \frac{7}{B} + 2 \times C + \frac{5}{D} + 6 \times E

then the value of 5*A* + *3*B + 6*C* + *D* + *3*E is :

Options:
- **(A)** `55.6003`
- **(B)** `53.603`
- **(C)** `153.6003`
- **(D)** `213.003`

Correct Answer: (C) `153.6003`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
4\times A + \frac{7}{B} + 2\times C + \frac{5}{D} + 6\times E =47.2506

=&gt; 4\times A + \frac{7}{B} + 2\times C + \frac{5}{D} + 6\times E = 40 + 7 + 0.2 + 0.05 + 0.0006

Comparing the terms on both sides, we get :

4*A =40, \frac{7}{B} = 7, 2*C = 0.2, \frac{5}{D} = 0.05, 6*E = 0.0006

Therefore:- A = 10, B = 1, C = 0.1, D = 100, E = 0.0001.

5*A + 3*B + 6*C + D + 3*E =(5*10) + (3*1) + (6*0.1) +100 +(3*0.0001)

= 50 + 3 + 0.6 + 100 + 0.003 = 153.6003.
```

Question 3 (PrepInsta quiz-1 - Q3)

Question Stem:

8. 3889 + 12.952 –? = 3854.002

Options:

- **(A)** `47.015`
- **(B)** `47.641`
- **(C)** `47.943`
- **(D)** `47.95`

Correct Answer: (D) `47.95`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Explanation:

Let 3889 + 12.952 - x = 3854.002.

Then x = (3889 + 12.952) - 3854.002

= 3901.952 - 3854.002

= 47.95
```

Question 4 (PrepInsta quiz-1 - Q4)

Question Stem:
83.04 x 2.01 x 2.1 = ?

Options:
- **(A)** `350.58124`
- **(B)** `333.8208`
- **(C)** `33.38208`
- **(D)** `35.152284`

Correct Answer: (C) `33.38208`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Given expression is, 83.04 x 2.01 x 2.1


The multiplication value without decimal point = 8304 x 201 x 2 = 3338208


Sum of the number of decimal places in the given numbers = 2 + 2 + 1 = 5


Now, in the product, the decimal point is marked off to obtain as many places of decimal.


Then the answer is 33.38208.
```

Question 5 (PrepInsta quiz-1 - Q5)

Question Stem:
By how much does 7/(8/9) exceeds (7/8)/9?

Options:
- **(A)** `2 (5/8)`
- **(B)** `7 (7/9)`
- **(C)** `3 (1/9)`
- **(D)** `9 (1/7)`

Correct Answer: (B) `7 (7/9)`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Here, we have to find the value of 7/(8/9) - (7/8)/9.

(7 x 9/8) - [(7/(8 x 9)]

63/8 - 7/72

[(63 x 9) - 7] / 72

567 - 7/72 = 560/72 = 70/9 = 7 7/9.

Hence, 7/(8/9) is greater than (7/8)/9 by 7 (7/9).
```

Question 6 (PrepInsta quiz-1 - Q6)

Question Stem:
The sum of the numerator and denominator of a fraction is 13. If we add 3 and 9 to the numerator and denominator respectively, it becomes 2/3. The fraction is:

Options:
- **(A)** `5/8`
- **(B)** `8/5`
- **(C)** `6/7`
- **(D)** `7/6`

Correct Answer: (D) `7/6`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Let the required fraction be p/q.

Given that, sum of the numerator and denominator of a fraction is 13;

i.e., p + q = 13 ...(1)

Adding 3 to numerator and 9 to the denominator, we have,

(P+3)/(q+9) = 2/3

3(P+3) = 2(q+9)

3p+9 = 2q+18

3p-2q = 9 ...(2)

Multiply eqn (1)by 2, 2p + 2q = 26...(3)

Adding (2) and (3),

5p = 35

p = 7.

Putting p value in (1), we get q = 13 - p = 13 - 7 = 6.

Hence the required fraction is p/q = 7/6
```

Question 7 (PrepInsta quiz-1 - Q7)

Question Stem:

What smallest fraction should be added to

$3\frac{2}{3} + 6\frac{7}{12} + 4\frac{9}{36} + 5 + 7\frac{1}{12}$ to make the sum a whole number?

Options:

- **(A)** `7/13`
- **(B)** `6/19`
- **(C)** `7/12`
- **(D)** `5/12`

Correct Answer: (D) `5/12`

PrepInsta Detailed Explanation & Mathematical Derivation:

$$3\frac{2}{3} + 6\frac{7}{12} + 4\frac{9}{36} + 5 + 7\frac{1}{12}$$

$$\text{Add only the fraction parts: } \frac{2}{3} + \frac{7}{12} + \frac{9}{36} + \frac{1}{12}$$

$$=\frac{24 + 21 + 9 + 3 }{36} = \frac{57}{36} = \frac{19}{12} = 1\frac{7}{12}$$

$$\text{Thus to make the whole number we should add } 1 - \frac{7}{12} = \frac{5}{12}$$

Question 8 (PrepInsta quiz-1 - Q8)

Question Stem:
In the CAPF Personality Interview a question was asked from a group of appearing candidates to check their analytical skills. The question is: if the sum of the two numbers is 70 and their product is 1221 then find the numbers.

Options:
- **(A)** `33,37`
- **(B)** `35,35`
- **(C)** `39,31`
- **(D)** `None of the above`

Correct Answer: (A) `33,37`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Let a and b are the numbers.Then a+b is 70 and ab is 1221.

a = 70 - b…... (i)

ab = 1221

(70-b)b = 1221

70b-b2-1221 = 0

Multiplying the above equation by (-1)

Then, the equation becomes: b2 - 70b + 1221 = 0

Solving the above equation we get through the middle term splitting method

b = 37

Substituting the value of b in equation (i) a = 33
```

Question 9 (PrepInsta quiz-1 - Q9)

Question Stem:
One day in Rahul's class their mathematics teacher challenged the whole class that if they will solve a question given by her she will give a free period to the class. The question was "What is the greatest of 5 consecutive integers whose sum is 215 ?". Help the class in finding the answer.

Options:
- **(A)** `42`
- **(B)** `43`
- **(C)** `44`
- **(D)** `45`

Correct Answer: (D) `45`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
The sum five consecutive integers can be written as n + (n + 1) + (n + 2)+(n+3)+(n+4) = 5n + 10

If the sum is 24, we need to solve the equation 5n+10 = 215;

=&gt; 5n = 205;

=&gt; n = 41

The greatest of 5 consecutive integers is n+4 = 41+4 = 45
```

Question 10 (PrepInsta quiz-1 - Q10)

Question Stem:

If the sum and difference of two numbers are 15 and 3 respectively. Then the difference of their square is?

Options:

- **(A)** `276`
- **(B)** `216`
- **(C)** `176`
- **(D)** `188`

Correct Answer: (B) `216`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Let the nos be “a” and “b”.

According to question,

a+b=15

and, a-b=3

 a^{2} - b^{2} = (a+b)(a-b)

 a^{2} - b^{2} = (15+3)(15-3)

 a^{2} - b^{2} = (18)(12)

 a^{2} - b^{2} = 216
```
